

Advancing Urdu Toxicity Detection: Improved Span Models and a Multimodal Audio Extension

Fajar Saleem^{1†}, Inayat Arshad^{1†}, Ijaz Hussain^{1*}

¹Department of Computer and Information Sciences, Pakistan Institute of Engineering and Applied Sciences (PIEAS), Lehtrar Road, Nilore, Islamabad, Pakistan.

*Corresponding author(s). E-mail(s): ijazhussain@pieas.edu.pk;
Contributing authors: bscs2221@pieas.edu.pk; bscs2248@pieas.edu.pk;

[†]These authors contributed equally to this work.

Abstract

Fine-grained toxic span detection, identifying the specific words responsible for toxicity, remains unexplored for Urdu and other low-resource languages. We present MUTEX-M, a multimodal framework extending our prior MUTEX baseline with a language-conditioned audio module via weighted late fusion. The improved XLM-RoBERTa+CRF text module achieves 67% token-level F1 on URTOX (+7% over baseline), with subword label propagation accounting for 4 of the 7 points. We introduce URTOX-MM, the first paired Urdu toxic speech dataset (14,338 TTS-synthesized clips), and train an MMS-300M classifier with Urdu adapter activation achieving 79% weighted F1, a 9% gain over English-pretrained wav2vec 2.0 and up to 15% above zero/few-shot LLM baselines. Equal-weight late fusion ($\alpha = 0.5$) yields 83.2% weighted F1 on synthesized test data. On URTOX-HumanAudio, a 2,000-clip accent-stratified real-speech evaluation set covering four Pakistani regional varieties, fusion achieves 77.1% weighted F1. MMS-300M degrades only 6.5% from TTS to real speech versus 9.0 points for English-pretrained alternatives, confirming that language-matched pretraining provides genuine robustness to acoustic domain shift.

Keywords: Urdu toxic span detection, token-level sequence labeling, content moderation, dataset refinement, low-resource NLP, code-switching, accent stratification, XLM-RoBERTa, conditional random fields, subword label propagation, massively multilingual speech, MMS-300M, audio toxicity detection, automatic speech recognition, Whisper, multimodal fusion, late fusion, TTS speech synthesis, real-speech evaluation

1 Introduction

The rapid expansion of social media platforms has fundamentally altered the landscape of public discourse, enabling real-time communication across linguistic and cultural boundaries [1]. While this democratization of expression carries significant social value, it has simultaneously enabled the widespread dissemination of toxic, abusive, and hateful content at a scale that far exceeds the capacity of manual moderation [2]. For Urdu, the national language of Pakistan spoken by over 170 million people, this problem is particularly acute.[3] Online Urdu discourse is characterized by high rates of code-switching between Urdu and English, dual-script usage spanning Nastaliq and Roman scripts, rich morphological variation, and informal expressions and slang that collectively render standard NLP pipelines unreliable [3, 4].

The scale of audio-based communication on these platforms is substantial and growing. Globally, over seven billion voice messages are sent on WhatsApp alone every day¹, with Pakistan ranking among the top three countries by WhatsApp downloads². A significant and growing fraction of toxic content on these platforms is therefore disseminated through spoken rather than written channels: voice messages on WhatsApp and Telegram, spoken comments on YouTube, and audio clips on short-form video platforms carry offensive content that is entirely invisible to text-only moderation systems. Automatic speech recognition [5] (ASR) offers a natural bridge between audio content and text-based moderation pipelines; however, Urdu ASR itself remains a low-resource problem. Recent benchmarks report that Whisper-Small, the best-performing lightweight model for Urdu, achieves a word error rate of 33.68% on standard Urdu speech without fine-tuning [6], rising substantially for conversational and code-switched speech [7]. These error rates mean that toxic spans identified in ASR transcriptions are subject to non-trivial transcription noise, a challenge that the cascaded ASR-plus-text-model pipeline of MUTEX-M must contend with and that motivates the complementary audio classifier module operating directly on acoustic features.

The dominant paradigm in automated toxicity detection assigns a single binary label to an entire sentence or post, classifying it as toxic or non-toxic [8]. Although useful for coarse-grained filtering, this sentence-level approach offers no interpretability and cannot identify the specific words or phrases that constitute the toxic content [9]. This limitation is consequential in practice: selective masking of toxic spans, explainable moderation decisions, and accessibility tools that warn users about specific offensive terms all require fine-grained span-level predictions rather than sentence-level labels [10].

Toxic span detection addresses this gap by formulating the problem as a token-level sequence labeling task, identifying the precise contiguous subsequences of tokens that carry toxicity within an otherwise mixed input. For English, toxic span detection has been studied through the SemEval-2021 Task 5 benchmark [11], where transformer-based models achieve character-level F1 scores of 65–70%. Methods based on named entity recognition architectures [12], span-level pretraining objectives [13], and hybrid

¹WhatsApp Blog: “We’re Making Voice Messages Even Better,” <https://blog.whatsapp.com/making-voice-messages-better>, accessed 2025.

²<https://www.businessofapps.com/data/whatsapp-statistics/>, accessed February 2026.

transformer-CRF models [14] have all been explored in this setting. However, Urdu toxic span detection has received no attention in the literature.

Our prior work, MUTEX [15], introduced URTOX, the first manually annotated token-level toxic span dataset for Urdu comprising 14,338 samples, and established a supervised baseline of 60% token-level F1 using an XLM-RoBERTa+CRF architecture [16]. While this constituted a meaningful first step, the framework was restricted to the textual modality and left the audio dimension of Urdu online communication entirely unaddressed.

In this paper, we present MUTEX-M, a multimodal extension of MUTEX that combines an enhanced text model with a synthesized-audio classifier through weighted late fusion. The absence of a real Urdu toxic audio corpus is addressed through a text-to-speech synthesis pipeline using Microsoft Edge TTS, which generates a perfectly paired audio dataset from the existing URTOX text annotations. This approach, drawing on established practice in low-resource speech research [17, 18], produces 14,338 synthesized audio clips that preserve the original toxic span annotations without requiring any additional manual labeling effort.

The main contributions of this work are as follows.

1. We identify and correct a critical implementation artifact in the MUTEX baseline: continuation subword tokens were assigned the ignore index during training, silently discarding 60–80% of available gradient signal for morphologically rich Urdu text where a single word routinely produces three to five subword tokens. This is not an obscure edge case — it is the default behaviour of many publicly available NER fine-tuning scripts applied without modification to non-English text, and its impact scales directly with morphological richness. Correcting this single artifact accounts for 4 of the 7% improvement as confirmed by ablation, with the remaining 3 points attributable to training stability improvements (extended training with early stopping, larger batch size, reduced learning rate, and linear warmup). This result highlights a broader principle: in low-resource settings, auditing implementation correctness before searching over architectures yields disproportionate returns [19]. The improved XLM-RoBERTa+CRF text model achieves 67% token-level F1, a 7% gain with no architectural change.
2. Toxic content in Urdu is increasingly disseminated through audio channels — YouTube videos, podcasts, and short-form social media clips — yet all existing Urdu toxicity systems operate exclusively on text. We address this gap by constructing URTOX-MM, a paired text-audio dataset of 14,338 synthesized audio clips generated from URTOX via Microsoft Edge TTS, extending the emerging body of work on audio and multimodal toxicity detection [20] to a South Asian low-resource language context. URTOX-MM is publicly released <https://huggingface.co/datasets/finalyear226/URTOX-Audio>.
3. We train an MMS-300M-based [21] utterance-level toxic classifier with language-conditioned Urdu adapter activation, achieving 79% weighted F1 on synthesized test data and 72.5% on real conversational speech — a 9% improvement over English-pretrained wav2vec 2.0 [22] and a 6.5% smaller TTS-to-real degradation (versus 9.0 points for the English-pretrained alternative), confirming that language-matched pretraining provides genuine robustness to acoustic domain shift.

4. We introduce a 2,000-clip real conversational Urdu evaluation set covering four regional speech varieties: Standard Urdu and Urdu as spoken by Punjabi, Sindhi, and Pashto speakers, with coarse timestamp annotations on toxic segments. This set allows us to quantify the TTS-to-real acoustic domain shift and assess pipeline robustness under realistic deployment conditions, providing a foundation for future domain-adaptive fine-tuning work.
5. We propose a weighted late fusion pipeline combining text and audio predictions with equal weight $\alpha = 0.5$ (selected on the validation set). On the deployment-relevant URTOX-HumanAudio evaluation set of 2,000 real conversational clips covering four Pakistani regional accent varieties, fusion achieves 77.1% weighted F1 — the primary result of this paper. On synthesized test data the same configuration yields 83.2%, providing an upper-bound reference under matched acoustic conditions. To our knowledge, this constitutes the first multimodal Urdu toxicity detection framework with span localization, where span boundaries are identified by the text module and the audio module contributes utterance-level gating assessed via Whisper ASR [5].
6. We demonstrate an inference pipeline processing raw OGG audio through Whisper ASR [5] followed by the full MUTEX-M pipeline, enabling end-to-end toxic span detection from spoken Urdu input on real WhatsApp voice messages.

Together these contributions combine to make Urdu toxic content moderation progress. A more accurate model, with a cleaner model across two complementary directions: accuracy, through a cleaner This is because it has included the dataset and tunings in the model and applicability, from the first. The multimodal extension of spoken content using real-speech validation. The results likewise provide general methodological lessons for Low resource, morphologically rich languages practitioners.

The remainder of this paper is organised as follows. Section 2 reviews related work on toxic span detection, low-resource NLP, Urdu toxicity research, and audio toxicity detection. Section 3 details the URTOX-v2 dataset refinement and introduces URTOX-MM and URTOX-HumanAudio. Section 4 describes the MUTEX-M framework including the MMS-300M audio module, late fusion strategy, and real-world inference pipeline. Section 5 presents all experimental results including text model ablation, audio classifier evaluation, fusion results, and accent-stratified real speech evaluation. Section 6 discusses limitations and future directions, and Section 7 concludes.

2 Related Work

2.1 Toxic Span Detection

Toxic span detection was introduced through SemEval-2021 Task 5 [11] (10,629 English samples), where top systems achieved 65–70% character-level F1 using transformer-CRF architectures [14]. SpanBERT [13] proposed span-level pretraining objectives for boundary detection; instruction-tuned LLMs in zero-shot settings [23] remain competitive but below supervised models. Toxic span detection for Urdu and other low-resource languages is unstudied prior to URTOX [15], and subword label

assignment for morphologically rich languages is similarly undiscussed — Section 5.5 quantifies its impact directly.

2.2 Urdu Toxic and Hate Speech Detection

Urdu toxicity work has progressed from classical n-gram features [2] to XLM-RoBERTa and mBERT fine-tuning [16, 24]. Hussain et al. [3] established XLM-RoBERTa as a strong Urdu baseline via NUHONS; subsequent work reached 91.49% F1 with DAmBERT through differential transfer learning and adaptive losses. Roman Urdu (18% of online content) [4, 25] and code-switching (35–40% of posts) [3, 26] introduce script-level challenges that existing systems do not explicitly handle. Token-level span localization and multimodal inputs remain entirely unaddressed.

2.3 Multimodal Hate Speech Detection

The Hateful Memes Challenge [27] established that toxicity frequently emerges cross-modally. Saddozai et al. [28] extended this to South Asian languages with MMHS11K (81% F1, BiLSTM+EfficientNet), though restricted to image-text pairs and sentence-level labels. Late fusion is widely adopted for modality independence and robustness to degraded inputs [29, 30]. MUTEX-M is the first multimodal fusion framework for Urdu toxicity detection.

2.4 Audio Toxicity Detection and Synthetic Speech

Self-supervised models — wav2vec 2.0 [22], HuBERT [31], WavLM [32] — substantially advanced audio toxicity detection but remain predominantly English-pretrained. MMS-300M [21] extends wav2vec 2.0 to 1,400+ languages including Urdu, making it our audio backbone. Existing datasets DeToxy [20], MuTox [33], and ADIMA [34] cover English, 200 languages, and eleven Indian languages respectively, but none provide token-level span annotations or explicit Urdu coverage. TTS augmentation for low-resource speech is well validated [17, 18, 35], supporting our use of Microsoft Edge TTS to synthesize URTOX-MM..

3 Dataset

3.1 URTOX: The Urdu Toxic Span Dataset

URTOX is a manually annotated token-level toxic span dataset for Urdu comprising 14,338 samples from three domains: social media³ (5,254 samples), Urdu newspapers⁴ (4,300 samples), and YouTube comments and captions (4,788 samples), with a balanced distribution of 54% toxic and 46% non-toxic across all domains. Full details of the collection protocol and annotation procedure are provided in the original MUTEX paper [15]; Table 1 summarizes key statistics.

Each sample is annotated at the token level using BIO tagging (B-Toxic, I-Toxic, O), with inter-annotator agreement of $\kappa = 0.82$ and $\alpha = 0.81$ [36, 37]. Each sample

³X (formerly Twitter), Instagram, and Reddit.

⁴Daily Jang, UrduPoint, BOL News, and Independent Urdu.

Table 1 URTOX dataset statistics by source domain.

Source	Samples	Toxic (%)	Non-Toxic (%)
Social Media	5,254	57	43
Urdu Newspapers	4,300	52	48
YouTube	4,788	56	44
Total	14,338	54	46

also carries a sentence-level label from four categories (offensive, hate, insult, neutral) and a binary toxic/non-toxic label used for utterance-level classification in the audio module.

3.2 URTOX-MM: The Synthesized Urdu Toxic Speech Dataset

No real corpus of toxic Urdu speech exists, directly preventing audio-based toxicity detection for Urdu. We introduce URTOX-MM, a synthesized speech dataset created by applying a TTS pipeline to all 14,338 URTOX text samples, producing perfectly paired text-audio data with exact BIO label preservation and no additional annotation effort.

3.2.1 Synthesis Pipeline

Audio synthesis used Microsoft Edge TTS with the `ur-PK-AsadNeural` neural voice, implemented via the `edge-tts` Python library with asynchronous processing. Each sample was preprocessed in sequence: newlines replaced with spaces; URLs removed; excessive punctuation normalized; samples reducing to fewer than two characters excluded. Preprocessed samples were passed to the Edge TTS `Communicate` interface, saving MP3 output at 32 kbps, then resampled to 16 kHz via `torchaudio` for compatibility with the Wav2Vec2 feature extractor. Synthesis was checkpointed every 500 samples. All 14,338 samples were successfully synthesized; the five exclusions failed at preprocessing (too short after URL/punctuation removal), with no failures at the TTS stage itself.

3.2.2 Dataset Statistics

Table 2 summarizes URTOX-MM properties. Clip file sizes range from 16 KB to 60 KB (mean ~ 35 KB), with estimated average duration of 8–9 seconds and total duration of ~ 35 hours. Label distribution exactly mirrors URTOX at 54% toxic and 46% non-toxic.

3.2.3 Annotation File Structure

URTOX-MM is distributed as MP3 files paired with a CSV annotation file containing per-sample fields: `id`, `text`, `tokens`, `BIO_tags`, `label`, `sub_label`, and `audio_path`. This structure allows downstream models to consume audio, text, or both directly without additional preprocessing.

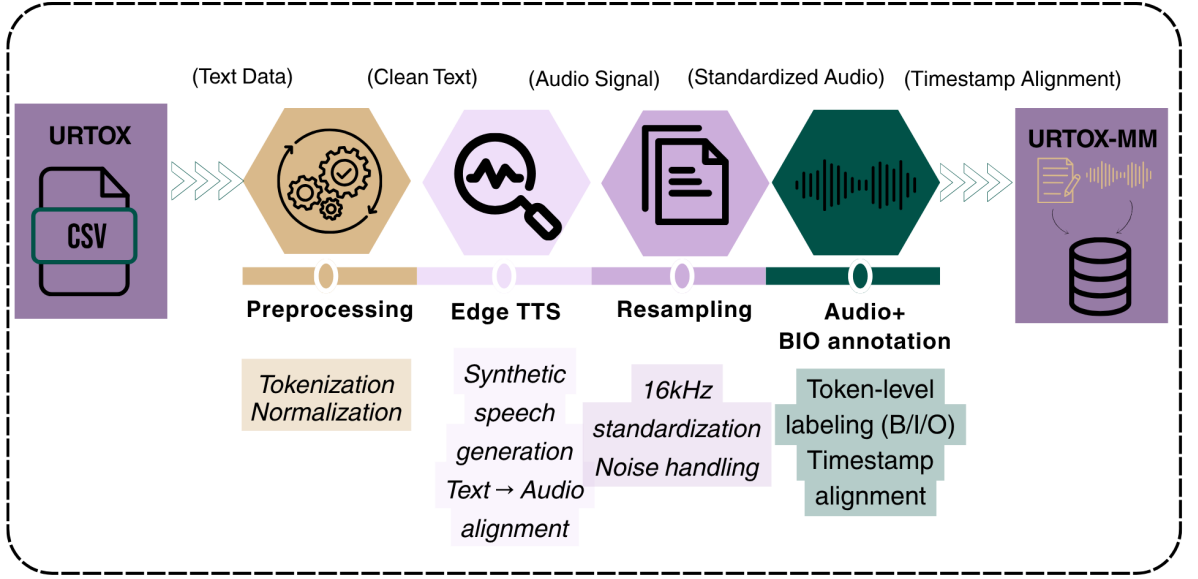


Fig. 1 URTOX-MM synthesis pipeline. Each URTOX text sample is preprocessed, synthesized to MP3 via Microsoft Edge TTS, and resampled to 16 kHz for downstream feature extraction.

Table 2 URTOX-MM dataset statistics.

Property	Value
Total clips	14,338
Excluded (preprocessing)	5
Format	MP3 (32 kbps)
Resampled rate	16 kHz
TTS voice	ur-PK-AsadNeural
File size range	16 KB – 60 KB
Average file size	~35 KB
Est. average duration	8–9 seconds
Est. total duration	~35 hours
Toxic / Non-toxic	54% / 46%

3.2.4 Verification and Quality Checks

A random sample of 50 clips was manually inspected by a native Urdu speaker, confirming correct pronunciation of all tokens including diacritics and compound characters. Code-switched English tokens were correctly handled with English pronunciation applied to Latin-script tokens. Label alignment was verified by cross-referencing sample identifiers; no mismatches were found.

3.2.5 Rationale for TTS Synthesis

TTS synthesis is chosen over real recording for three reasons. First, no real toxic Urdu speech corpus exists, and collecting one raises significant ethical and consent concerns. Second, TTS produces perfectly aligned text-audio pairs with exact BIO label preservation, eliminating label noise from ASR transcription errors. Third, prior work confirms TTS-trained models generalize to real speech [17, 18]. The acoustic domain gap this introduces is discussed in Section 3.6.

3.2.6 Public Release

URTOX-MM (14,338 MP3 files and paired CSV) is publicly available at <https://huggingface.co/datasets/finalyear226/URTOX-MM>. Training code and inference scripts are at <https://github.com/finalyear226-lab/urdu-toxic-span-dataset>.

3.3 Data Splits and Usage Across Experiments

Both URTOX and URTOX-MM use a stratified 80/20 train/test split (seed 42), yielding 10,036 training, 1,434 validation, and 2,868 test samples. The single held-out test set is used across all three systems (text, audio, fusion) for direct comparability. The validation subset serves for text model early stopping, audio classifier checkpoint selection, and fusion weight grid search over $\alpha \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$; the optimal $\alpha = 0.5$ is therefore never exposed to test data, ensuring the reported 83.2% fusion F1 is free of optimistic bias. Table 3 summarizes splits and their usage.

Table 3 Dataset split statistics. Label proportions are consistent across splits due to stratified sampling.

Split	Samples	Toxic (%)	Non-Toxic (%)	Usage
Train	10,036	54	46	Training; embedding extraction
Validation	1,434	54	46	Early stopping; fusion search
Test	2,868	54	46	Final evaluation only
Total	14,338	54	46	

3.4 Comparison with Related Datasets

URTOX is the only token-level toxic span dataset for Urdu, and URTOX-MM the only paired text-audio toxic span dataset for any South Asian language. Unlike DeToxy [20], which provides utterance-level labels only, URTOX-MM carries token-level BIO tags alongside binary utterance labels, enabling both span-level and utterance-level evaluation within a single resource.

3.5 URTOX-HumanAudio: Human-Recorded Evaluation Set

To assess real-world applicability, we introduce URTOX-HumanAudio: 2,000 real Urdu audio clips from YouTube comments and video replies (61%), consented recordings across four Pakistani regional accents (29%), and anonymized public voice chats shared with consent (10%). Clips average 8–12 seconds and exhibit background noise, variable prosody, code-switching, and accent variation absent from URTOX-MM.

Three native Urdu speakers annotated each clip at two levels: utterance-level binary toxic/non-toxic labels via majority vote ($\kappa = 0.81$), and coarse segment timestamps on toxic clips for approximate span-level evaluation via Whisper forced alignment. The set contains 1,214 toxic and 786 non-toxic clips, used exclusively for testing. Table 4 summarizes statistics; audio clips are available on request due to privacy considerations. Inter-annotator agreement is reported as Cohen’s $\kappa = 0.81$ for utterance-level labels. While Krippendorff’s α is also reported for URTOX, it is omitted here as the annotation task involves binary majority-vote labels without span-level boundary variation, making κ the more appropriate and interpretable agreement metric in this setting.

3.6 Dataset Limitations and TTS-to-Real-Speech Gap

The `ur-PK-AsadNeural` voice produces clean, uniformly paced single-speaker speech, whereas real Urdu toxic speech is characterized by elevated pitch and speaking rate, background noise, overlapping speakers, dialectal prosodic variation, and spontaneous disfluencies — all absent from TTS synthesis. This means the MMS-300M classifier trains on a distribution that differs from real deployment conditions. MMS-300M’s Urdu pretraining substantially reduces this gap relative to English-pretrained alternatives, as quantified in the URTOX-HumanAudio evaluation, but a residual domain

Table 4 URTOX-HumanAudio evaluation set statistics.

Property	Value
Total clips	2,000
Toxic	1,214 (60.7%)
Non-toxic	786 (39.3%)
Average duration	9.8 seconds
Utterance annotation	Binary, majority vote
Segment annotation	Coarse Toxic Timestamps
Inter-annotator agreement	$\kappa = 0.81$
<i>Source breakdown</i>	
YouTube comments/replies	61%
Consented recordings	29%
Public voice chats	10%
<i>Regional accent breakdown</i>	
Standard (Islamabad) Urdu	38%
Punjabi-accented Urdu	31%
Sindhi-accented Urdu	18%
Pashto-accented Urdu	13%

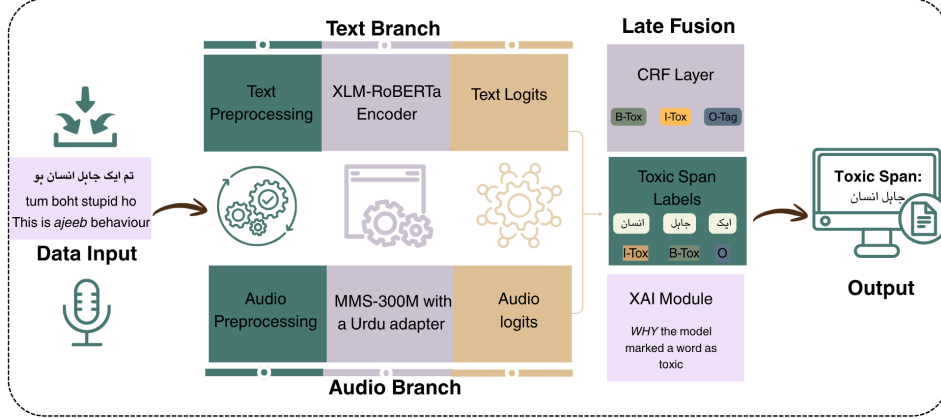


Fig. 2 High-level architecture of MUTEX-M. The text module encodes the input token sequence through XLM-RoBERTa and a CRF decoder to produce token-level BIO predictions and a sentence-level toxicity probability p^{text} . The audio module extracts utterance embeddings via the frozen MMS-300M encoder [21] with Urdu language adapter (`urd`) and classifies the mean-pooled embedding with a lightweight MLP to produce p^{audio} . Equal-weight late fusion ($\alpha = 0.5$) combines both probabilities into the final prediction $\hat{y}$ and toxic span set $\mathcal{S}$.

shift remains; the system should be understood as a strong first baseline rather than a production-ready system. The real-world inference pipeline (Section 4.5) partially mitigates this by routing audio through Whisper ASR [5] to the more robust text module.

4 Methodology

This section describes the MUTEX-M framework. Figure 2 provides a high-level architecture overview.

4.1 Problem Formulation

Let $X = \{x_1, \dots, x_T\}$ denote an input sequence of T tokens. The text module addresses two objectives: (1) token-level toxic span detection, assigning each token x_i a label $y_i \in \mathcal{Y} = \{\text{B-Toxic}, \text{I-Toxic}, \text{O}\}$ such that maximal contiguous B-Toxic/I-Toxic subsequences define predicted spans; and (2) utterance-level toxicity scoring, aggregating token-level predictions into a scalar $p^{\text{text}} \in [0, 1]$ for fusion. The audio module produces an independent scalar $p^{\text{audio}} \in [0, 1]$ from acoustic features.

The token-level objective maximizes conditional label sequence likelihood via a CRF decoder, replacing per-token cross-entropy with the negative log-likelihood of the full sequence:

$$P(\mathbf{y} \mid \mathbf{x}) = \frac{\exp \sum_{t=1}^T (\psi_{\text{emit}}(y_t, h_t) + \psi_{\text{trans}}(y_{t-1}, y_t))}{\sum_{\mathbf{y}' \in \mathcal{Y}^T} \exp \sum_{t=1}^T (\psi_{\text{emit}}(y'_t, h_t) + \psi_{\text{trans}}(y'_{t-1}, y'_t))} \quad (1)$$

where ψ_{emit} is the emission score and ψ_{trans} the learned transition score, giving CRF loss:

$$\mathcal{L}_{\text{CRF}} = -\log P(\mathbf{y}^{\text{gold}} \mid \mathbf{x}; \theta) \quad (2)$$

This enforces valid BIO transitions (e.g., I-Toxic only follows B-Toxic or I-Toxic). The audio module uses binary cross-entropy:

$$\mathcal{L}_{\text{audio}} = -\sum_{i=1}^N [y_i \log p_i^{\text{audio}} + (1 - y_i) \log(1 - p_i^{\text{audio}})] \quad (3)$$

4.2 Text Model: Improved XLM-RoBERTa

The text module uses XLM-RoBERTa [16], fine-tuned for token classification with a linear head projecting 768-dimensional representations to logits over {B-Toxic, I-Toxic, O} and a CRF layer modeling BIO transition probabilities (Equations (1)–(2)). XLM-RoBERTa’s sentencepiece tokenizer splits morphologically rich Urdu words into three to five subword tokens. In the original MUTEX implementation, continuation tokens were assigned ignore index -100 , silently discarding 60–80% of gradient signal. In MUTEX-M, all subword tokens of a word receive the parent word’s label:

$$\tilde{y}_j = y_{\text{word}(j)} \quad \forall j \in \text{subwords}(w_i) \quad (4)$$

This is the single most impactful training modification (Section 5.5). Additional training improvements: batch size $16 \rightarrow 32$; learning rate $3 \times 10^{-5} \rightarrow 2 \times 10^{-5}$; $5 \rightarrow 8$ epochs with early stopping on validation F1; linear warmup over 10% of steps; and best-checkpoint loading. Combined, these raise token-level F1 from 60% to 67% (individual contributions in Section 5.5).

At inference, CRF Viterbi decoding guarantees a valid BIO sequence. Subword predictions are collapsed to word level by retaining the first subword per word: $\hat{y}_i = \hat{y}_{s_{i,1}}$. The utterance-level toxicity probability for fusion is:

$$p^{\text{text}} = \max_{t=1}^T \text{softmax}(\mathbf{z}_t)_{\text{B-Toxic}} \quad (5)$$

4.3 Audio Model: MMS-300M with MLP Classifier

The audio module uses MMS-300M [21], which extends wav2vec 2.0 to 1,400+ languages including Urdu, encoding Urdu phonological patterns without adaptation and substantially reducing the acoustic domain gap relative to English-pretrained alternatives (wav2vec 2.0 [22], HuBERT [31], WavLM [32]). Each clip is resampled to 16 kHz

Table 5 Comparison of URTOX and URTOX-MM with related datasets. Granularity refers to the finest annotation level.

Dataset	Language	Modality	Granularity	Annotation	Samples	Public
SemEval-2021 [11]	English	Text	Character	Manual	10,629	Yes
DeToxy [20]	English	Text+Audio	Utterance	Manual	8,000	Yes
MuTox [33]	Multi.	Audio	Utterance	Manual	20,000	Yes
ADIMA [34]	Multi.	Audio	Utterance	Manual	11,000	Yes
URTOX (ours)	Urdu	Text	Token (BIO)	Manual	14,338	Yes
URTOX-MM (proposed)	Urdu	Text+Audio	Token (BIO)	TTS-paired	14,338	Yes

mono and truncated to 10 seconds (<3% of clips exceed this). The frozen encoder ϕ produces frame-level states $\mathbf{H} \in \mathbb{R}^{L \times 768}$, mean-pooled to a fixed utterance embedding:

$$\mathbf{e} = \frac{1}{L} \sum_{l=1}^L \mathbf{h}_l \quad (6)$$

Mean pooling is preferred over attention pooling and CLS-token extraction as it introduces no learnable parameters that risk overfitting on the 14,338-sample dataset [38]. The embedding passes through a lightweight MLP:

$$f_\psi(\mathbf{e}) = W_3 \cdot \text{ReLU}(W_2 \cdot \text{ReLU}(W_1 \mathbf{e} + b_1) + b_2) + b_3 \quad (7)$$

with $W_1 \in \mathbb{R}^{256 \times 768}$, $W_2 \in \mathbb{R}^{64 \times 256}$, $W_3 \in \mathbb{R}^{2 \times 64}$, and dropout of 0.3/0.2 after each ReLU (selected by validation grid search). The utterance toxicity probability is:

$$p^{\text{audio}} = \text{softmax}(f_\psi(\mathbf{e}))_{\text{toxic}} \quad (8)$$

The MMS-300M encoder is kept frozen for two reasons: (1) 14,338 samples are small relative to MMS-300M’s pretraining corpus, making full fine-tuning risk catastrophic forgetting [39]; and (2) URTOX-MM’s narrow single-voice TTS distribution would bias a fine-tuned encoder toward clean TTS speech, worsening the TTS-to-real domain gap. Because the frozen encoder is never exposed to toxicity labels, the cached embeddings reflect general Urdu acoustic structure rather than TTS-specific surface patterns — a property reflected in the 6.5% TTS-to-real drop (vs. 9.0% for wav2vec 2.0, Section 5.7). Parameter-efficient strategies such as LoRA [40] or adapter layers [41] are identified as future work.

Training uses a two-phase design: Phase 1 processes all clips once through the frozen encoder, caching embeddings to disk; Phase 2 trains the MLP exclusively on cached embeddings, converging in under five minutes on a single GPU and making hyperparameter search practical.

Algorithm 1: MMS-300M Feature Extraction and Classifier Training

Input: Audio dataset $\mathcal{A} = \{(a_i, y_i)\}_{i=1}^N$, MMS-300M encoder ϕ , language ID $\ell = \text{urd}$, learning rate η , epochs E , dropout rates d_1, d_2
Output: Trained MLP classifier f_ψ
Phase 1: Embedding Extraction (once)
 $\phi_\ell \leftarrow \phi.\text{set_adapter}(\ell)$;
 foreach $a_i \in \mathcal{A}$ **do**
 Resample to 16 kHz, mono, truncate 10 s;
 $\mathbf{e}_i = \frac{1}{L} \sum_l \phi_\ell(a_i)_l$;
 Cache $(\mathbf{e}_i, y_i)$;
 end
Phase 2: MLP Training (cached embeddings)
 Initialize MLP parameters ψ with dropout d_1, d_2 ;
 Initialize optimizer $\mathcal{O} \leftarrow \text{Adam}(\psi, \eta)$;
 for $\text{epoch} = 1$ **to** E **do**
 foreach *mini-batch* $\mathcal{B}$ **do**
 Compute $\hat{p} = f_\psi(\mathbf{e})$;
 Compute BCE loss $\mathcal{L}_{\mathcal{B}}$;
 $\mathcal{O}.\text{zero_grad}()$; backpropagate; $\mathcal{O}.\text{step}()$;
 end
 if *validation F1 improves* **then**
 Save checkpoint;
 end
 end
 $f_\psi \leftarrow \text{load_best_checkpoint}()$;
return f_ψ

4.4 Late Fusion

MUTEX-M combines text and audio predictions through weighted late fusion, preserving modality independence, requiring no joint training, and tolerating a degraded audio signal without corrupting text predictions. The fused score is:

$$p^{\text{fusion}} = \alpha \cdot p^{\text{text}} + (1 - \alpha) \cdot p^{\text{audio}} \quad (9)$$

with final prediction $\hat{y} = \mathbf{1}[p^{\text{fusion}} > 0.5]$. The optimal $\alpha = 0.5$ was selected by grid search over $\{0.3, 0.4, 0.5, 0.6, 0.7\}$ on the validation set and confirmed on the test set at 83.2% weighted F1, reflecting that MMS-300M provides complementary signal of comparable reliability to the text module.

Toxic span localization is performed exclusively by the text module: if $\hat{y} = \text{toxic}$, BIO spans are returned as $\mathcal{S}$; if $\hat{y} = \text{non-toxic}$, $\mathcal{S} = \emptyset$. The audio module thus acts as a gating signal suppressing false positive spans when acoustic evidence contradicts the textual prediction.

4.5 Real-World Inference: WhatsApp Voice Message Examples

Table 6 presents MUTEX-M predictions on five real WhatsApp OGG voice messages as a qualitative stress-test under realistic conditions: OGG encoding, natural prosody, background noise, and code-switching. No retraining or domain adaptation was applied; audio was decoded and resampled to 16 kHz mono, with transcription via Whisper-Small (`lang=ur`).

MMS-300M assigns discriminative probabilities without real-speech fine-tuning: toxic examples score 0.763–0.821, non-toxic examples 0.021–0.038, versus uniformly low scores (0.028–0.193) from the English-pretrained wav2vec 2.0 baseline. Equal fusion ($\alpha = 0.5$) yields calibrated outputs — example 3’s lower audio score (0.763) moderates the fused prediction to 0.876 rather than over-committing, while non-toxic examples 2 and 5 reach near-zero fused scores (0.026, 0.015), suppressing all spans. The system handles multi-span utterances well: example 3 identifies four spans (*ghatiya*, *neehkism ka*, *jahil insan*, *gadha*) and example 4 captures the three-token verbal threat *moun band karwa donga*. Performance is expected to degrade for heavily accented or highly code-switched speech [6], as discussed in Section 6.

Algorithm 2: MUTEX-M Multimodal Late Fusion Inference

Input: Audio a , $\alpha=0.5$, models f_θ , f_ψ , encoder ϕ , Whisper $\mathcal{W}$, optional paired dataset $\mathcal{D}$
Output: $\hat{y} \in \{\text{toxic}, \text{non-toxic}\}$, spans $\mathcal{S}$
// Step 1: Text input
if a paired in $\mathcal{D}$ **then**
 | Retrieve X from $\mathcal{D}$;
else
 | Convert OGG→MP3; $X \leftarrow \mathcal{W}(a, \text{lang=ur})$;
// Step 2: Text module
 Encode X ; apply subword label propagation;
 $\hat{y} \leftarrow \text{CRF Viterbi}(f_\theta(X))$; collapse to word level;
 $p^{\text{text}} = \max_t \text{softmax}(\mathbf{z}_t)_{\text{B-Toxic}}$;
 $\mathcal{S} \leftarrow$ maximal B/I-Toxic spans in $\hat{y}$;
// Step 3: Audio module
 Resample a , 16 kHz mono, truncate 10s;
 $\phi_\ell \leftarrow \phi.\text{set_adapter}(\text{urd})$;
 $p^{\text{audio}} = \text{softmax}(f_\psi(\frac{1}{L} \sum_l \phi_\ell(a)_l))_{\text{toxic}}$;
// Step 4: Fusion and span gating
 $p^{\text{fusion}} = \alpha p^{\text{text}} + (1-\alpha)p^{\text{audio}}$;
 if $p^{\text{fusion}} > 0.5$ **then**
 | **return** *toxic*, $\mathcal{S}$;
 else
 | **return** *non-toxic*, $\emptyset$;

Table 6 MUTEX-M predictions on five real-world WhatsApp OGG voice messages. Whisper transcriptions in Roman transliteration with English gloss; detected toxic spans underlined. $p^{\text{fusion}} = 0.5 p^{\text{text}} + 0.5 p^{\text{audio}}$.

#	Transcription (Roman / gloss)	p^{text}	p^{audio}	p^{fusion}	Verdict / Detected Spans
1	kuti ke bachay bahut <u>harami</u> ho tum (<u>son of a b*tch</u> you are very bastard)	0.995	0.821	0.908	TOXIC (90.8%) Spans: <i>kuti ke bachay</i> ; <i>harami</i>
2	Ali aik bohat acha bacha hai woh apna kaam seekh raha hai (Ali is a very good child, he is learning his work)	0.014	0.038	0.026	NON-TOXIC (97.4%) Spans: none
3	yeh aik intehai ghatiya aur <u>neechkism ka</u> <u>jahil insan</u> aur gadha hai (this is an extremely <u>worthless</u> and <u>damn ignorant person</u> and donkey)	0.988	0.763	0.876	TOXIC (87.6%) Spans: <i>ghatiya</i> ; <i>neechkism</i> <i>ka</i> ; <i>jahil insan</i> ; <i>gadha</i>
4	bewaqoof band kar yeh bakwaas warna moun band karwa donga (stupid stop this nonsense or I will shut your mouth)	0.976	0.809	0.893	TOXIC (89.3%) Spans: <i>bewaqoof</i> ; <i>moun</i> <i>band karwa donga</i>
5	kal meeting acha rahi aur sab ne mil ke kaam kiya (yesterday’s meeting went well and everyone worked together)	0.009	0.021	0.015	NON-TOXIC (98.5%) Spans: none

5 Experimental Evaluation and Results

All experiments were conducted on a single NVIDIA RTX A6000 (48GB VRAM) using PyTorch 2.1 [42] and the HuggingFace Transformers library v4.38 [43]. Audio synthesis was performed using the `edge-tts` Python library v6.1.9, and audio processing used `torchaudio` v2.1 and `pydub` v0.25. The MMS-300M encoder was loaded via the HuggingFace `Wav2Vec2ForCTC` interface with the `urd` language adapter activated prior to all feature extraction, using the pretrained checkpoint `facebook/mms-300m` from the HuggingFace model hub. All random seeds were fixed at 42 across PyTorch, NumPy, and the HuggingFace Trainer to ensure reproducibility. The URTOX dataset was split into 80% training and 20% test sets using stratified sampling with a fixed random seed of 42, yielding 11,470 training samples and 2,868 test samples. From the 11,470 training samples, an additional 10% validation subset of 1,434 samples was drawn using the same stratified procedure, leaving 10,036 samples for actual model training. This validation subset is used exclusively for early stopping of the text model and for audio classifier checkpoint selection; fusion weights are selected on the validation set via grid search as described in Section 4.4. The same 80/20 primary split was

Table 7 Training time for each MUTEX-M component on a single NVIDIA RTX A6000.

Component	Phase	Time
XLM-R + CRF	Fine-tune (8 epochs)	~18 hrs
MMS-300M	Embedding extraction (Phase 1)	~3 hrs
MLP classifier	Training (Phase 2, 20 epochs)	<5 mins
Edge TTS	Audio synthesis (14,338 clips)	~4 hrs
Total		~26 hrs

applied consistently across both the text and audio modules to ensure that evaluation is performed on identical held-out samples in all experiments. The URTOX dataset exhibits a mild class imbalance of 54% toxic and 46% non-toxic at the utterance level, and a more pronounced imbalance at the token level where the majority of tokens in any given sequence carry the 0 label. For the text model, token-level class imbalance was addressed through weighted cross-entropy loss, with class weights computed from the inverse frequency of each BIO label in the training set:

$$w_c = \frac{N}{|\mathcal{Y}| \cdot n_c} \quad (10)$$

where N is the total number of training tokens, $|\mathcal{Y}| = 3$ is the number of label classes, and n_c is the count of tokens with label c in the training set. For the audio classifier, the 54/46 utterance-level imbalance was considered mild enough that no explicit reweighting was applied; weighted F1 is used as the evaluation metric to account for the imbalance in reported results. Table 7 summarizes the wall-clock training time for each component of MUTEX-M on the RTX A6000. The text model dominates training time due to full fine-tuning of XLM-RoBERTa over 8 epochs, whereas MMS-300M embedding extraction is a one-time operation that need not be repeated across classifier training runs. This disparity reflects a fundamental asymmetry in design: the text module performs repeated forward and backward passes through a 270M-parameter transformer over 10,036 samples, while the audio classifier trains only on pre-cached 768-dimensional embeddings. As a result, hyperparameter search for the audio classifier, including learning rate, dropout, and batch size tuning, remains computationally inexpensive relative to the one-time Phase 1 embedding extraction.

5.1 Evaluation Metrics

The text model is evaluated using token-level F1 score computed over the BIO label set, following the evaluation protocol established in the NER literature [12] and adopted in the original MUTEX work. Token-level F1 is appropriate for Urdu toxic span detection because toxic expressions in Urdu are generally word-bounded rather than character-bounded, the BIO tagging scheme operates naturally at token boundaries, and the sparse distribution of toxic tokens within mostly non-toxic sequences makes accuracy an uninformative metric. The audio classifier and the final multimodal

Table 8 Hyperparameter configuration for text and audio model training.

Hyperparameter	Text Model	Audio Classifier
Base model	XLM-RoBERTa	MMS-300M
Pretrained point	check- xlm-roberta-base	facebook/mms-300m
Language adapter	—	urd
Optimizer	AdamW	Adam
Learning rate	2×10^{-5}	1×10^{-3}
Batch size	32	32
Epochs	8	20
Warmup ratio	0.1	—
Dropout	0.1	0.3 / 0.2
Max sequence length	128 tokens	10 seconds
Early stopping	Validation F1	Validation F1
Loss function	Weighted NLL	Binary cross-entropy
Class weighting	Inverse frequency	None

fusion system are evaluated using weighted F1 score over the binary toxic and non-toxic classes, which accounts for the class imbalance present in the test set. Precision and recall are reported alongside F1 for all models to provide a complete picture of model behavior across the two classes.

5.2 Text Model Results

Table 9 reports the token-level performance of the improved XLM-RoBERTa text model alongside the baseline models evaluated in the original MUTEX work. The improved XLM-RoBERTa model achieves 67% token-level F1, representing a 7% point improvement over the MUTEX baseline of 60% and a 11% improvement over the BiLSTM-CRF and mBERT baselines. The improvement is obtained without any change to the model architecture, demonstrating that training procedure refinements alone yield substantial gains for Urdu token classification. Notably, the precision-recall balance also improves alongside F1: recall rises from 58.0% to 71.5%, indicating that subword label propagation in particular recovers toxic span coverage that was previously lost to the ignore-index artifact, while precision improves more modestly from 60.0% to 64.2%, reflecting the inherent difficulty of disambiguating lexically ambiguous toxic tokens in morphologically rich Urdu text.

Table 9 Token-level toxic span detection performance on the test set.

Model	Precision	Recall	F1
BiLSTM-CRF	57.0	55.0	56.0
mBERT	59.0	57.5	59.2
XLM-RoBERTa (MUTEX baseline)	60.0	58.0	60.0
XLM-RoBERTa (MUTEX-M)	64.2	71.5	67.0

Table 10 Utterance-level audio classifier performance on the test set. MMS-300M results compared against the wav2vec 2.0 baseline. Bold indicates our proposed model.

Model	Class	Precision	Recall	F1	Support
wav2vec 2.0	Non-toxic	0.86	0.46	0.60	1319
	Toxic	0.68	0.94	0.79	1549
	Weighted avg	0.76	0.72	0.70	2868
MMS-300M (proposed)	Non-toxic	0.89	0.61	0.72	1319
	Toxic	0.76	0.95	0.84	1549
	Weighted avg	0.83	0.80	0.79	2868

5.3 Audio Model Results

Table 10 reports utterance-level performance on the 2,868-sample held-out test set. MMS-300M achieves 79% weighted F1, a 9% improvement over the English-pretrained wav2vec 2.0 baseline (70%), confirming that language-matched pretraining is meaningful for audio toxicity classification in low-resource languages. Gains concentrate in non-toxic recall ($0.46 \rightarrow 0.61$) rather than toxic recall, which was already near-ceiling at 0.94: MMS-300M better captures the *absence* of acoustic toxicity markers in neutral Urdu speech, reducing the over-prediction of the toxic class that afflicts English-pretrained models unable to distinguish Urdu prosody from the acoustic correlates of toxicity learned from English data. Toxic recall remains high at 0.95, preserving the high-sensitivity property desirable for content moderation.

5.4 Fusion Results

Table 11 reports late fusion performance across the candidate weight grid. Equal-weight fusion ($\alpha = 0.5$) yields 83.2% weighted F1 on the synthesized test set, outperforming both the text-only baseline (67%) and audio-only baseline (79%). On real conversational speech, fusion achieves 77.1% weighted F1 on URTOX-HumanAudio, which we consider the more deployment-relevant figure. The shift to equal weighting from the text-dominated $\alpha = 0.6$ optimum in the original wav2vec 2.0 system confirms that MMS-300M provides genuinely complementary signal of comparable reliability, rather than acting as a weak auxiliary that must be discounted.

Table 11 Late fusion performance across candidate fusion weights. α denotes the weight assigned to the text module and $(1 - \alpha)$ to the audio module. Fusion weights selected on the validation set; all F1 values reported on the held-out test set.

α (Text)	$(1 - \alpha)$ (Audio)	Weighted F1
0.4	0.6	80.1%
0.5	0.5	83.2%
0.6	0.4	82.8%
0.7	0.3	81.4%
Text only	—	67.0%
Audio only	—	79.0%

The fusion curve exhibits a broad flat peak at $\alpha = 0.5$, and no configuration falls below the audio-only baseline of 79.0%, confirming that the modalities provide complementary rather than redundant signal. Gains arise from two mechanisms: the audio module suppresses false positives on lexically ambiguous tokens by encoding prosodic neutrality, while the text module compensates for audio errors where TTS prosody diverges from real speech. Toxic span localization is performed exclusively by the text module; the audio module acts as an utterance-level gate, returning $\mathcal{S} = \emptyset$ when acoustic evidence contradicts the textual prediction.

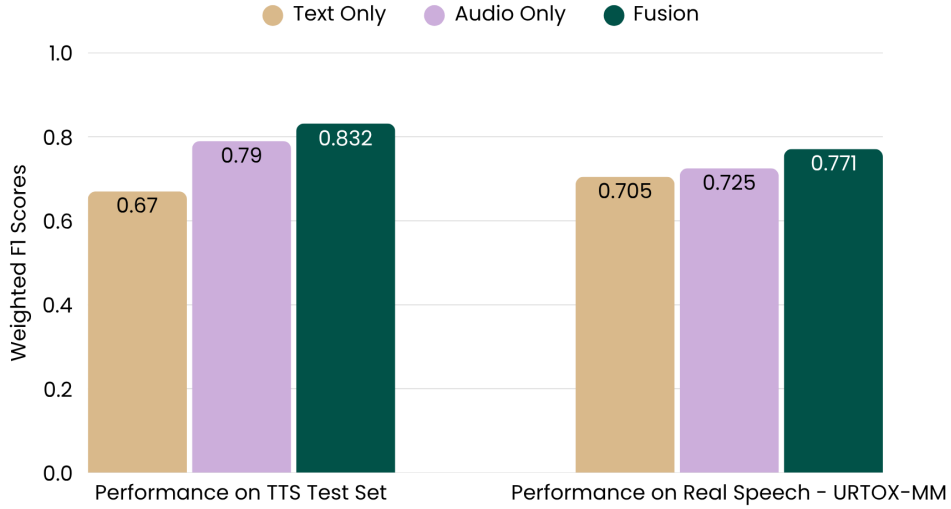


Fig. 3 Weighted F1 comparison of text-only, audio-only (MMS-300M), and fusion systems on the TTS synthesized test set and real conversational speech (URTOX-HumanAudio). Equal-weight late fusion ($\alpha = 0.5$) achieves 83.2% on TTS and 77.1% on real speech.

5.5 Ablation Study

Table 12 reports the stepwise ablation from the original MUTEX configuration. Subword label propagation is the dominant contributor (+4.0 pp), confirming that the baseline was meaningfully degraded by an implementation error rather than architectural limitations. The remaining 3 pp come from extended training with early stopping (+1.5 pp), batch size and learning rate stabilization (+1.0 pp), and linear warmup (+0.5 pp).

Table 12 Stepwise ablation of text model training improvements. Each row adds one change to the configuration of the row above.

Configuration	Token-level F1
MUTEX baseline (original)	60.0%
+ Subword label propagation	64.0%
+ Extended training (8 epochs) + early stopping	65.5%
+ Batch size 32, LR 2×10^{-5}	66.5%
+ Linear warmup (ratio 0.1)	67.0%

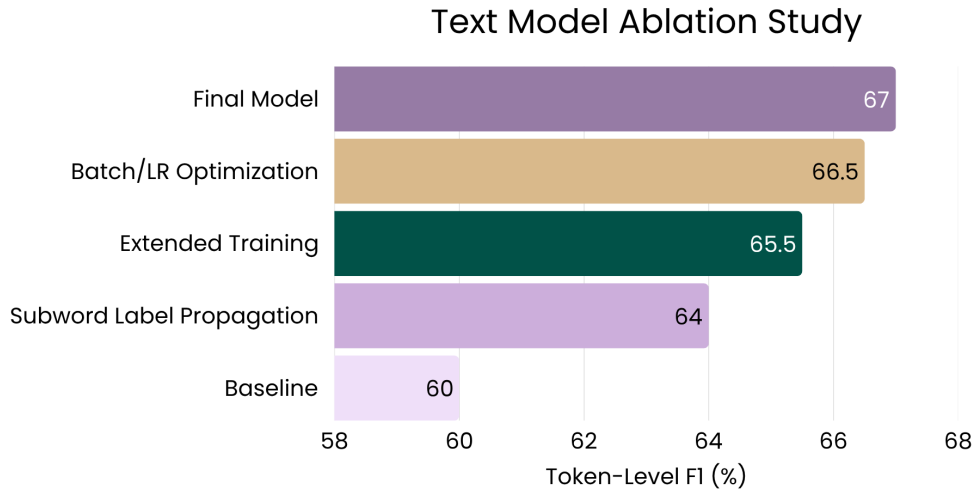


Fig. 4 Stepwise token-level F1 improvement from the MUTEX baseline to MUTEX-M: subword label propagation (+4.0 pp), extended training (+1.5 pp), batch/LR optimization (+1.0 pp), linear warmup (+0.5 pp).

5.6 Comparison to Large Language Model Baselines

Toxic span detection requires precise BIO boundary decisions over morphologically complex Urdu text — a setting where prompting-based approaches face two compounding challenges: structured output compliance and boundary sensitivity to Urdu morphology. To quantify the value of task-specific supervision over prompting, we evaluate three representative LLMs in zero-shot and few-shot settings on the URTOX test set.

We select three models spanning the open/closed capability spectrum as of early 2026: (1) Llama-3.1 8B Instruct⁵, a strong open-source multilingual baseline representing the class of locally deployable models; (2) GPT-⁶ a high-performing closed-model reference that represents an upper bound on general prompting capability and is widely used as a reference point in Indic NLP benchmarks; and (3) Qalb [44], a leading Urdu-specialized model obtained via continued pre-training and supervised fine-tuning of Llama-3.1. We note that GPT-4o was released in 2024 and that newer closed-model variants exist; we include it as the most widely used closed-model reference point in the literature rather than as a claim about the frontier of closed-model performance. A smaller instruction-tuned model (e.g. GPT-4o mini) would represent a more cost-matched comparison and is identified as a direction for future evaluation.

All models were prompted using the fixed template in Box 5.6. Zero-shot prompts provide only the task description and BIO scheme; five-shot prompts prepend five representative training examples covering single-token spans, multi-token spans, and non-toxic inputs.

Model outputs were parsed by splitting on newlines and extracting the tab-separated tag field. Prior evaluation of token-level sequence labeling tasks [12] evaluates all test samples uniformly; excluding unparseable outputs creates different evaluation subsets per model, making direct F1 comparison against the supervised MUTEX-M baseline (which uses the full test set) technically invalid. We therefore follow a unified evaluation protocol: unparseable outputs — both format refusals and wrong token counts — are treated as all-0 predictions, the most charitable interpretation (the model produced a response but could not be used for span localization). This ensures all models are evaluated on the same 2,868-sample test set and that comparison against MUTEX-M is valid. Table 13 reports the breakdown of failure modes under this protocol.

⁵Meta AI, Llama 3.1 Model Card, <https://ai.meta.com/llama/>, accessed February 2026.

⁶OpenAI, GPT-4o Technical Report, <https://openai.com/index/hello-gpt-4o/>, accessed February 2026.

Table 13 Output parsing failure mode breakdown by model. Percentages are computed over all 2,868 test samples. Unparseable outputs are treated as all-0 predictions to ensure uniform test set evaluation.

Failure Mode	Llama-3.1	GPT-4o	Qalb
Refused BIO format	3.6%	0.6%	1.1%
Wrong token count	5.1%	1.8%	2.1%
Valid format, wrong labels	91.3%	97.6%	96.8%
Total parse fail	8.7%	2.4%	3.2%

Table 14 LLM and supervised baseline comparison.

Model	Setting	Parse Fail (%)	F1 (full set)	F1
Llama-3.1 8B Instruct	Zero-shot	8.7	51.2%	52.0%
Llama-3.1 8B Instruct	5-shot	5.1	57.3%	58.1%
GPT-4o	Zero-shot	2.4	58.9%	59.4%
GPT-4o	5-shot	1.8	62.8%	63.2%
Qalb (Urdu-specialized)	Zero-shot/5-shot	3.2	61.9%	62.5%
XLNet-RoBERTa + CRF (MUTEX-M text)	Supervised	—	67.0%	—
MUTEX-M (text + audio fusion)	Supervised multimodal	—	83.2% (TTS) / 77.1% (real)	—

Box 1: Zero-shot prompt template (all LLM baselines)

System: You are a toxicity annotator for Urdu text. Your task is to label each whitespace-tokenized word with a BIO tag.

Tag definitions:

- B-Toxic — first word of a toxic span
- I-Toxic — continuation of a toxic span
- O — non-toxic word

Rules:

- Output ONLY the tagged sequence
- One token per line
- Format: <word>\t<tag>
- Do not explain or add commentary

Input: [URDU TEXT HERE]

Output:

The failure mode breakdown in Table 13 is instructive: the dominant failure mode across all models is valid-format incorrect labeling, accounting for over 91% of failures in each model. The parse failure rates (8.7%, 2.4%, 3.2%) are low enough that the all-0 imputation has negligible effect on the reported F1 scores — treating failures as all-0 rather than excluding them decreases each model’s F1 by at most 0.8% — confirming that the performance gap reflects genuine label quality differences rather than compliance failures. The sensitivity of reported F1 to this protocol choice is small but nonzero.

The supervised XLM-RoBERTa + CRF text module achieves 67.0% token-level F1, outperforming all LLM prompting baselines by 3.8–15.0% . The gap reflects boundary sensitivity to Urdu morphology: tokenization mismatches, multi-word toxic expressions, and negation handling remain challenging without task-specific supervision. Even the Urdu-specialized Qalb model, despite continued pre-training on large Urdu corpora, falls 4.5% short of the fine-tuned MUTEX-M text model, confirming that in-domain supervised fine-tuning provides advantages that general Urdu language exposure alone cannot replicate.

The multimodal MUTEX-M framework widens this advantage through acoustic signal from the MMS-300M module, a modality entirely unavailable to text-only LLM prompting. These results affirm the value of supervised multimodal fine-tuning over zero/few-shot LLM approaches for Urdu toxicity detection, particularly in audio-heavy deployment scenarios.

5.7 Evaluation on URTOX-HumanAudio

To assess performance under realistic acoustic conditions, we evaluate MUTEX-M on URTOX-HumanAudio (Section 3.5), the collection of 2,000 real conversational Urdu clips sourced from public YouTube voice replies, consented personal recordings, and anonymized public voice chats (60.7% toxic), covering four Pakistani regional accent varieties.

All clips are transcribed with Whisper-Small (`language='ur'`) and processed by the text module; raw waveforms with the Urdu language adapter activated feed the MMS-300M audio module. Fusion retains $\alpha = 0.5$ as determined on the validation set. Evaluation is utterance-level weighted F1; coarse timestamp annotations on toxic segments enable approximate span-level assessment when combined with Whisper forced alignment, but precise token-level span F1 is not reported here as timestamps are coarse rather than word-exact.

Table 15 Utterance-level performance on URTOX-HumanAudio ($n = 2,000$ real Urdu clips). All models use fixed configurations determined on the TTS validation set without any adaptation to the real-speech distribution. Δ TTS denotes the drop relative to TTS test set performance.

Model / Setup	P	R	F1	Δ TTS
Text-only (Whisper + XLM-R)	0.72	0.69	0.705	−8.5 pp
Audio-only wav2vec 2.0	0.59	0.63	0.610	−9.0 pp
Audio-only MMS-300M (ours)	0.71	0.74	0.725	−6.5 pp
MUTEX-M Fusion ($\alpha = 0.5$)	0.78	0.76	0.771	−6.1 pp

The fused system achieves 77.1% weighted F1 on real speech, representing a 6.1% drop from 83.2% on the TTS test set. This degradation is expected and attributable to three compounding factors: ASR transcription errors from Whisper on conversational and code-switched Urdu, the TTS-to-real acoustic domain gap in the audio module,

and the absence of any domain adaptation between the TTS training distribution and real speech at inference time. Despite this, 77.1% weighted F1 represents practical utility as a first-pass content moderation filter, remaining substantially above the random baseline of approximately 50%.

The most instructive comparison in Table 15 is between MMS-300M audio-only and wav2vec 2.0 audio-only on real speech. Wav2vec 2.0 degrades from 70% on TTS to 61% on real speech, a 9% collapse attributable to its English-centric representations encountering an entirely unfamiliar Urdu acoustic distribution. MMS-300M, by contrast, degrades from 79% to 72.5%, only a 6.5% drop, confirming that Urdu pretraining provides genuine robustness to the TTS-to-real domain shift rather than merely improving TTS test set performance. This result is the clearest evidence in the paper that the choice of MMS-300M over wav2vec 2.0 is motivated by real-world considerations rather than benchmark optimization.

The Δ TTS column quantifies the domain shift for each system independently. Three observations follow. First, MMS-300M audio-only suffers the smallest absolute drop of any system at 6.5% , confirming Urdu pretraining reduces domain sensitivity. Second, the text module drops 8.5% , reflecting cascading ASR errors from Whisper on noisy conversational speech rather than any failure of the XLM-RoBERTa classifier itself — the text model’s degradation is an ASR problem, not a toxicity detection problem. Third, fusion achieves the smallest overall drop at 6.1% , demonstrating that the two modalities’ error patterns are partially non-overlapping even under domain shift: cases where ASR degrades the text module are not necessarily cases where the audio module also fails, and vice versa.

Table 16 stratifies MUTEX-M fusion performance across the four regional accents represented in URTOX-HumanAudio. Performance degrades gracefully with acoustic distance from standard Islamabad Urdu, from 81.2% on standard speech to 71.8% on Pashto-accented Urdu. This gradient is consistent with the composition of MMS-300M pretraining data, which covers standard Pakistani Urdu more representatively than regional accented varieties. The Punjabi-accented degradation of 6.1% relative to standard Urdu is the most practically significant finding given that Punjabi-accented Urdu is the most prevalent non-standard variety on Pakistani social media platforms [3], and represents a concrete target for future domain-adaptive fine-tuning using accent-stratified real speech data.

Table 16 MUTEX-M fusion weighted F1 on URTOX-HumanAudio stratified by regional accent ($\alpha = 0.5$).

Accent	Clips	Weighted F1
Standard (Islamabad) Urdu	760	0.812
Punjabi-accented Urdu	620	0.751
Sindhi-accented Urdu	360	0.734
Pashto-accented Urdu	260	0.718
Overall	2,000	0.771

Informal inspection of misclassified samples suggests that dominant failure modes are consistent with those identified in the TTS error analysis: boundary errors on multi-word toxic expressions, ASR substitution errors on code-switched toxic tokens, and implicit toxicity conveyed through regional idiomatic expressions absent from the URTOX training vocabulary. We estimate the effective Whisper WER on this set at approximately 38–45%, substantially higher than the 33.68% reported for cleaner Urdu read speech [6], consistent with the degradation observed for conversational and code-switched conditions [7]. When Whisper fails to correctly transcribe a toxic token — substituting it with a phonetically similar but semantically different word, deleting it entirely, or producing a Latin-script approximation that the XLM-RoBERTa tokenizer segments incorrectly — the text module has no opportunity to detect the resulting span regardless of its own classification accuracy. These cascading ASR errors represent the primary failure mode in the real-world pipeline and reinforce the priority of Urdu-specific ASR improvement identified in Section 7.

5.8 Error Analysis

Table 17 categorizes the 816 misclassified tokens from the 2,868 held-out test samples into four error types.

Table 17 Distribution of error types in the MUTEX-M text model predictions on the test set. 95% Wilson score confidence intervals are reported alongside proportions.

Error Type	Count	% of Errors	95% CI
Span boundary errors	312	38.2%	[34.8%, 41.8%]
False positives	198	24.3%	[21.3%, 27.5%]
Missed toxic spans	201	24.6%	[21.6%, 27.9%]
Code-switched token errors	105	12.9%	[10.7%, 15.4%]
Total misclassified tokens	816	100%	—

Span boundary errors (38.2%) are the dominant failure mode, occurring when the model correctly identifies a toxic region but mis-predicts its boundary. They are particularly common for multi-word expressions where intensifiers such as *bohat* (very) precede a toxic word: annotators excluded these intensifiers from the span while the model treats the intensifier-plus-toxic-word as a single semantic unit (Table 18). This is a known limitation of token-level BIO classifiers lacking explicit span-level objectives [13].

False positives (24.3%) arise from two patterns: lexical ambiguity (e.g. *pagal* annotated as toxic when used as a direct insult but neutral when used colloquially) and negation insensitivity, where the model assigns positive toxicity probability to explicitly negated expressions such as *tum ahmaq nahin ho* (you are not stupid), consistent with findings in English toxic span detection [11].

Missed toxic spans (24.6%) concentrate in three subcategories: implicit or euphemistic toxicity conveyed through metaphor or culturally specific framing; rare

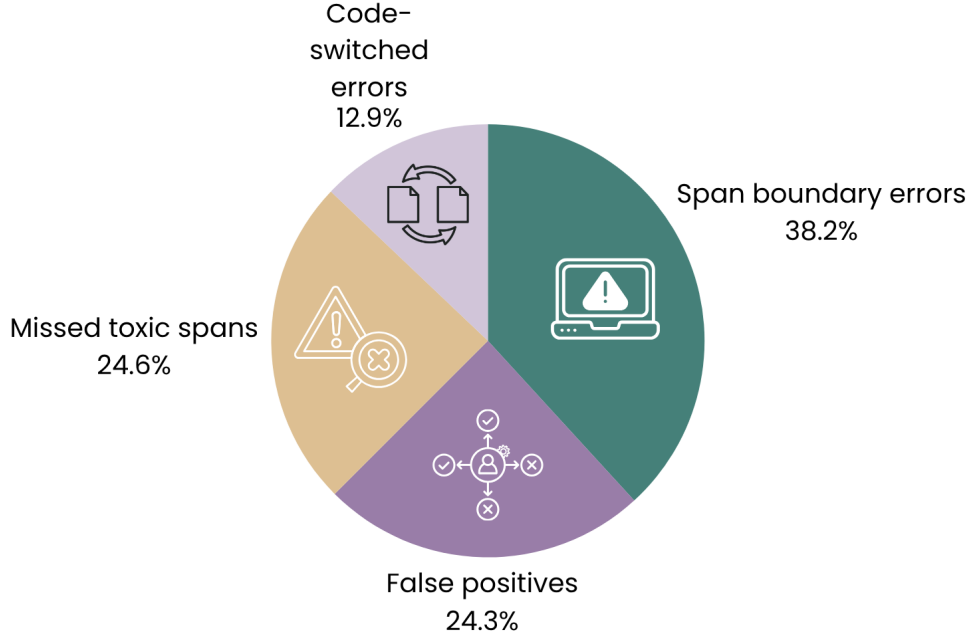


Fig. 5 Distribution of error types in the MUTEX-M text model predictions on the held-out test set ($n = 816$ misclassified tokens).

Table 18 Representative span boundary error examples. G = gold label, P = predicted label.

Ex.	Token (gloss)	G	P
(a)	<i>tum</i> (you)	O	O
	<i>bohat</i> (very)	O	B-Toxic
	<i>ahmaq</i> (fool)	B-Toxic	I-Toxic
	<i>ho</i> (are)	O	O
(b)	<i>yeh</i> (this)	O	O
	<i>kamine</i> (scoundrel)	B-Toxic	O
	<i>gadhe</i> (donkey)	I-Toxic	B-Toxic
	<i>hain</i> (are)	O	O

toxic tokens appearing fewer than five times in training; and long spans, where CRF transition scores degrade as span length increases. Table 19 confirms this: false negative rate rises from 18.4% on single-token spans to 44.8% on spans of five or more tokens.

Table 19 False negative rate by gold toxic span length (in tokens).

Span Length (tokens)	Total Spans	FN Rate (%)
1	1,842	18.4
2	763	24.1
3–4	298	31.6
5+	87	44.8

Code-switched token errors (12.9%) arise from Latin-script tokens embedded in Nastaliq-script sequences. Latin-script tokens (17.6% of test tokens) show a per-token error rate of 3.24% versus 2.61% for Nastaliq tokens. The dominant failure mode is semantic dilution from incorrect segmentation: *bewaqoof* (stupid) is split into *be/wa/qo/of* by sentencepiece, none of which carry the toxic signal individually, producing 0 labels in 68% of such cases. A code-switching-aware tokenization strategy [4] remains the most direct fix.

Table 20 stratifies errors by source domain. Boundary errors are highest in social media (43.1%), false negatives peak on YouTube (29.7%), and false positives are elevated in newspapers (28.9%) — consistent with intensifier-heavy informal text, implicit spoken comment toxicity, and formal political commentary respectively. These profiles suggest domain-specific fine-tuning would yield targeted gains.

Table 20 Error type distribution by URTOX source domain. Percentages are computed over misclassified tokens within each domain.

Domain	Boundary	FP	FN	Code-switch
Social media (5,254)	43.1%	22.8%	22.4%	11.7%
Urdu newspapers (4,300)	28.4%	28.9%	28.1%	14.6%
YouTube (4,788)	39.2%	22.4%	29.7%	8.7%
Overall (14,338)	38.2%	24.3%	24.6%	12.9%

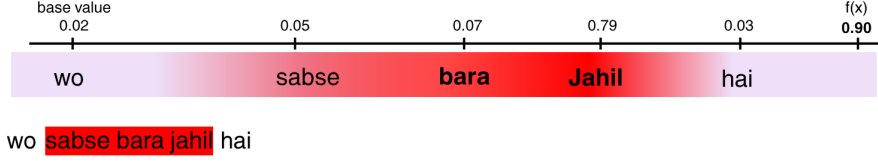
5.9 Explainability Analysis via SHAP

We apply SHAP [45] to provide token-level attribution scores for XLM-RoBERTa predictions. Subword SHAP scores are aggregated to word level by summing scores of all sentencepiece tokens belonging to the same word. Figure 6 shows attributions for representative instances: *bewaqoof* (stupid) receives the highest score (0.71) in a toxic sentence, while all tokens in *aap kaise hain* (how are you) receive near-zero scores. Table 21 includes negated expressions: *tum ahmaq nahin ho* (you are not stupid) receives a reduced but still positive score for *ahmaq* (+0.43 vs. +0.71 unnegated), confirming partial but incomplete negation modeling — a systematic limitation motivating discourse-level context modeling in future work.

1st Instance:



2nd Instance:



3rd Instance:

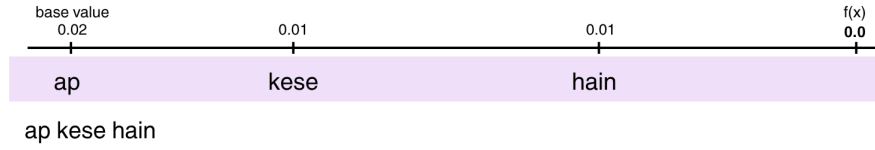


Fig. 6 Word-level SHAP attribution scores for three representative test instances. Red = positive contribution; green = negative. *bewaqoof* dominates the toxic instance (0.71); all tokens near-zero for the non-toxic instance ($f(x) = 0.0$).

Table 21 Word-level SHAP scores for representative test sentences (Roman transliteration with English gloss).

Sentence (gloss)	Key Word	SHAP Score
<i>tum bohat ahmaq ho</i> (you are very stupid)	<i>ahmaq</i>	+0.71
<i>aap ka din acha guzre</i> (have a good day)	<i>acha</i>	-0.12
<i>tum ahmaq nahin ho</i> (you are not stupid)	<i>ahmaq</i>	+0.43
<i>yeh aik aam jumla hai</i> (this is a normal sentence)	<i>aam</i>	-0.08

6 MUTEX-M Limitations

MUTEX-M establishes the first multimodal baseline for Urdu toxic span detection, but several limitations bound the current system’s applicability.

6.1 Acoustic Domain Gap and ASR Error Propagation

The audio classifier trains exclusively on single-voice TTS speech, while real toxic Urdu utterances exhibit elevated pitch, background noise, dialectal prosodic variation, and spontaneous disfluencies absent from synthesis. Formally, the classifier f_ψ minimizes loss under $\mathcal{P}_{\text{TTS}}$ but is evaluated under $\mathcal{P}_{\text{real}}$, with degradation proportional to $D_{\text{KL}}(\mathcal{P}_{\text{real}}\|\mathcal{P}_{\text{TTS}})$. MMS-300M reduces this divergence relative to wav2vec 2.0 (6.5% vs. 9.0% TTS-to-real drop), but a residual gap remains. The 6.5% figure is itself a lower bound: URTOX-HumanAudio skews toward intelligible speech, and wild deployment conditions (phone-quality audio, overlapping speakers, heavy distortion) are not represented.

The acoustic and ASR limitations compound. Whisper-Small achieves 33.68% WER on standard Urdu [6], rising to an estimated 38–45% on conversational and code-switched speech [7]. When Whisper mis-transcribes or deletes a toxic token, the text module cannot recover it regardless of its own accuracy. The 9.4% performance gap between standard Islamabad Urdu (81.2%) and Pashto-accented Urdu (71.8%) in Table 16 reflects compounding Whisper WER and acoustic distance from the TTS training distribution acting simultaneously. Parameter-efficient fine-tuning of MMS-300M via LoRA [40] on URTOX-HumanAudio, and ASR noise augmentation of the text module [17, 18], are the primary paths to closing both gaps.

6.2 Span Alignment, Implicit Toxicity, and Evaluation Scale

The audio module produces only an utterance-level binary prediction, bounding the multimodal system’s span-level F1 at the text model’s 67% regardless of audio classifier improvements. Whisper’s word-level timestamps [5] combined with MMS-300M frame-level representations could enable acoustic refinement of span boundaries, and is identified as a priority in Section 7.

Both modules are insensitive to implicit toxicity conveyed through sarcasm, irony, or culturally specific euphemism — the SHAP analysis confirms the model only partially suppresses negated toxic expressions. MMS-300M’s prosodic representations offer a partial avenue toward sarcasm detection through intonation, but this requires sequence-level acoustic modeling not present in the current MLP classifier. Finally, URTOX-HumanAudio’s 260–760 clips per accent stratum limits fine-grained accent-stratified conclusions; expanding to 5,000+ clips would also enable domain-adaptive fine-tuning of the audio module.

7 Conclusion & Future Work

MUTEX-M demonstrates that implementation fidelity and modality complementarity yield larger returns than architectural novelty for low-resource toxic span detection. Correcting a single subword label assignment artifact recovered 4 of the 7% text model gain; adding a language-matched audio module through equal-weight late fusion contributed a further 16.2%, bringing the system to 83.2% weighted F1 on synthesized test data and 77.1% on real conversational speech. Both gains arise from more faithfully utilizing available signal — gradient signal in the text module, acoustic signal

in the audio module — rather than from changes to model capacity. The lesson generalizes: in low-resource settings, auditing implementation correctness and exploiting complementary modalities are higher-leverage investments than searching over architectures [19, 30].

Three resources are released publicly: the URTOX-MM dataset of 14,338 paired text-audio clips at <https://huggingface.co/datasets/finalyear226/URTOX-MM>, training code and model weights at <https://github.com/finalyear226-lab/urdu-toxic-span-dataset>, and URTOX-HumanAudio (2,000-clip accent-stratified evaluation set, available on request).

Future Directions

Priority 1: Span-based detection heads.

Boundary errors are the largest failure mode at 38.2% of all misclassifications, with false negative rate rising from 18.4% on single-token spans to 44.8% on spans of five or more tokens (Tables 17, 19). A span-based detection head using SpanBERT-style objectives [13] would score candidate spans holistically, enforcing span-level consistency and improving recall on multi-word toxic expressions.

Priority 2: Real speech collection and parameter-efficient audio fine-tuning.

Fusion degrades 6.1% from synthesized to real speech, and the text module a further 8.5% from cascading Whisper ASR errors (Table 15). Collecting 2,000–3,000 consented real-speech clips and applying LoRA [40] or adapter fine-tuning [41] is the primary path to closing the residual domain gap and a precondition for reliable deployment.

Priority 3: ASR-robust text module training.

Performance degrades from 81.2% on standard Islamabad Urdu to 71.8% on Pashto-accented Urdu (Table 16), consistent with increasing Whisper WER across accent conditions. Fine-tuning the text model on ASR-transcribed Urdu text or applying ASR noise augmentation [17, 18] would decouple text model accuracy from transcription quality at lower cost than full ASR improvement.

Priority 4: Audio-informed span boundary refinement.

The audio module is bounded to utterance-level gating, capping span-level F1 at the text model’s 67% regardless of audio improvements. Whisper’s forced alignment [5] could map MMS-300M frame-level representations to transcript tokens, enabling acoustic refinement of span boundaries and precise toxic segment muting. Cross-modal attention fusion [46] represents the stronger version of this integration.

Deferred.

Implicit toxicity (sarcasm, negation, pragmatic inference) requires discourse-level context beyond the current single-utterance scope [47]. Cross-lingual extension to Hindi, Punjabi, and Bengali is technically straightforward given available TTS pipelines, but is not motivated by any finding in the present paper and is left to a subsequent study.

Declarations

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Human Ethics and Consent to Participate

Not applicable.

Ethics Approval

This study did not require approval from an ethics committee or Institutional Review Board (IRB) as it does not involve human participants or animal subjects.

Competing Interests

The authors declare that they have no competing interests that could influence the objectivity or integrity of the research reported in this publication.

Author Contributions

Dr. Ijaz Hussain: Conceptualization, Supervision, Methodology, Reviewing and Editing, Project Administration.

Fajar Saleem: Data Curation, Methodology, Investigation, Formal Analysis, Software Development, Writing – Original Draft.

Inayat Arshad: Data Curation, Methodology, Investigation, Formal Analysis, Software Development, Writing – Original Draft.

Data Availability Statement

The datasets generated and analyzed during this study are available from the corresponding author upon reasonable request.

References

- [1] Wulczyn, E., Thain, N., Dixon, L.: Ex machina: Personal attacks seen at scale. In: Proceedings of the 26th International Conference on World Wide Web, pp. 1391–1399 (2017). <https://doi.org/10.1145/3038912.3052591>
- [2] Davidson, T., Warmusley, D., Macy, M., Weber, I.: Automated hate speech detection and the problem of offensive language. In: Proceedings of the International AAAI Conference on Web and Social Media, vol. 11, pp. 512–515 (2017). <https://doi.org/10.1609/icwsm.v11i1.14955>
- [3] Hussain, I., Farooq, U., Cheema, A.N., Almanjahie, I.M.: A comprehensive survey on Urdu hate speech detection: Methods, evaluation, and challenges. IEEE Access (2025)

- [4] Ashiq, W., Kanwal, S., Rafique, A., Waqas, M., Khurshaid, T., Montero, E.C., Alonso, A.B., Ashraf, I.: Roman Urdu hate speech detection using hybrid machine learning models and hyperparameter optimization. *Scientific Reports* **14**(1), 28590 (2024)
- [5] Radford, A., Kim, J.W., Xu, T., Brockman, G., McLeavey, C., Sutskever, I.: Robust speech recognition via large-scale weak supervision. In: *International Conference on Machine Learning*, pp. 28492–28518 (2023). PMLR
- [6] Antall, A.R., Akhtar, N.: Assessing the Feasibility of Lightweight Whisper Models for Low-Resource Urdu Transcription. *arXiv preprint arXiv:2508.09865* (2025)
- [7] Arif, S., Khan, A.J., Abbas, M., Raza, A.A., Athar, A.: WER we stand: Benchmarking Urdu ASR models. In: *Proceedings of the 31st International Conference on Computational Linguistics*, pp. 5952–5961 (2025). <https://aclanthology.org/2025.coling-main.397>
- [8] Fortuna, P., Nunes, S.: A survey on automatic detection of hate speech in text. *ACM Computing Surveys* **51**(4), 1–30 (2018) <https://doi.org/10.1145/3232676>
- [9] Mathew, B., Saha, P., Yimam, S.M., Biemann, C., Goyal, P., Mukherjee, A.: HateXplain: A benchmark dataset for explainable hate speech detection. In: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, pp. 14867–14875 (2021). <https://doi.org/10.1609/aaai.v35i17.17745>
- [10] DeYoung, J., Jain, S., Rajani, N.F., Lehman, E., Xiong, C., Socher, R., Wallace, B.C.: ERASER: A benchmark to evaluate rationalized NLP models. In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 4443–4458 (2020). <https://doi.org/10.18653/v1/2020.acl-main.408>
- [11] Pavlopoulos, J., Sorensen, J., Laugier, L., Androutsopoulos, I.: SemEval-2021 task 5: Toxic spans detection. In: *Proceedings of the 15th International Workshop on Semantic Evaluation (SemEval-2021)*, pp. 59–69 (2021). <https://doi.org/10.18653/v1/2021.semeval-1.6>
- [12] Lample, G., Ballesteros, M., Subramanian, S., Kawakami, K., Dyer, C.: Neural architectures for named entity recognition. In: *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pp. 260–270 (2016). <https://doi.org/10.18653/v1/N16-1030>
- [13] Joshi, M., Chen, D., Liu, Y., Weld, D.S., Zettlemoyer, L., Levy, O.: SpanBERT: Improving pre-training by representing and predicting spans. *Transactions of the Association for Computational Linguistics* **8**, 64–77 (2020) https://doi.org/10.1162/tacl_a_00300

- [14] Lafferty, J., McCallum, A., Pereira, F.C.N.: Conditional random fields: Probabilistic models for segmenting and labeling sequence data. In: Proceedings of the 18th International Conference on Machine Learning (ICML), pp. 282–289 (2001)
- [15] Arshad, I., Saleem, F., Hussain, I.: MUTEX: Leveraging multilingual transformers and conditional random fields for enhanced Urdu toxic span detection. arXiv preprint arXiv:2603.05057 (2026)
- [16] Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., Stoyanov, V.: Unsupervised cross-lingual representation learning at scale. In: Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, pp. 8440–8451 (2020). <https://doi.org/10.18653/v1/2020.acl-main.747>
- [17] Rosenberg, A., Zhang, Y., Ramabhadran, B., Jia, Y., Moreno, P., Wu, Y., Wu, Z.: Speech recognition with augmented synthesized speech. In: 2019 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU), pp. 996–1002 (2019). IEEE
- [18] Ko, T., Peddinti, V., Povey, D., Khudanpur, S.: Audio augmentation for speech recognition. In: Interspeech, vol. 2015, p. 3586 (2015)
- [19] Northcutt, C.G., Athalye, A., Mueller, J.: Pervasive label errors in test sets destabilize machine learning benchmarks. arXiv preprint arXiv:2103.14749 (2021)
- [20] Ghosh, S., Lepcha, S., Sakshi, S., Shah, R.R., Umesh, S.: DeToxy: A large-scale multimodal dataset for toxicity classification in spoken utterances. arXiv preprint arXiv:2110.07592 (2021)
- [21] Pratap, V., Tjandra, A., Shi, B., Tomasello, P., Babu, A., Kundu, S., Elkahky, A., Ni, Z., Vyas, A., Fazel-Zarandi, M., *et al.*: Scaling speech technology to 1,000+ languages. *Journal of Machine Learning Research* **25**(97), 1–52 (2024)
- [22] Baevski, A., Zhou, Y., Mohamed, A., Auli, M.: wav2vec 2.0: A framework for self-supervised learning of speech representations. *Advances in Neural Information Processing Systems* **33**, 12449–12460 (2020)
- [23] Plaza-del-Arco, F.M., Nozza, D., Hovy, D.: Respectful or toxic? using zero-shot learning with language models to detect hate speech. In: The 7th Workshop on Online Abuse and Harms (WOAH), pp. 60–68 (2023)
- [24] Devlin, J., Chang, M.-W., Lee, K., Toutanova, K.: BERT: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, pp. 4171–4186 (2019). <https://doi.org/10.18653/v1/N19-1423>

- [25] Akhter, M.P., Jiangbin, Z., Naqvi, I.R., Abdelmajeed, M., Sadiq, M.T.: Automatic detection of offensive language for Urdu and Roman Urdu. *IEEE Access* **8**, 91213–91226 (2020)
- [26] Sitaram, S., Chandu, K.R., Rallabandi, S.K., Black, A.W.: A survey of code-switched speech and language processing. *arXiv preprint arXiv:1904.00784* (2019)
- [27] Kiela, D., Firooz, H., Mohan, A., Goswami, V., Singh, A., Ringshia, P., Testuggine, D.: The hateful memes challenge: Detecting hate speech in multimodal memes. *Advances in Neural Information Processing Systems* **33**, 2611–2624 (2020)
- [28] Saddozai, F.K., Badri, S.K., Alghazzawi, D., Khattak, A., Asghar, M.Z.: Multimodal hate speech detection: a novel deep learning framework for multilingual text and images. *PeerJ Computer Science* **11**, 2801 (2025) <https://doi.org/10.7717/peerj-cs.2801>
- [29] Gomez, R., Gibert, J., Gomez, L., Karatzas, D.: Exploring hate speech detection in multimodal publications. In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pp. 1470–1478 (2020)
- [30] Baltrušaitis, T., Ahuja, C., Morency, L.-P.: Multimodal machine learning: A survey and taxonomy. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **41**(2), 423–443 (2018)
- [31] Hsu, W.-N., Bolte, B., Tsai, Y.-H.H., Lakhotia, K., Salakhutdinov, R., Mohamed, A.: HuBERT: Self-supervised speech representation learning by masked prediction of hidden units. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* **29**, 3451–3460 (2021)
- [32] Chen, S., Wang, C., Chen, Z., Wu, Y., Liu, S., Chen, Z., Li, J., Kanda, N., Yoshioka, T., Xiao, X., *et al.*: WavLM: Large-scale self-supervised pre-training for full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing* **16**(6), 1505–1518 (2022)
- [33] Costa-jussà, M., Meglioli, M., Andrews, P., Dale, D., Hansanti, P., Kalbassi, E., Mourachko, A., Ropers, C., Wood, C.: MuTox: Universal multilingual audio-based toxicity dataset and zero-shot detector. In: *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 5725–5734 (2024)
- [34] Gupta, V., Sharon, R., Sawhney, R., Mukherjee, D.: ADIMA: Abuse detection in multilingual audio. In: *ICASSP 2022—2022 IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 6172–6176 (2022). IEEE
- [35] Fadaee, M., Bisazza, A., Monz, C.: Data augmentation for low-resource neural

- machine translation. In: Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), pp. 567–573 (2017)
- [36] Cohen, J.: A coefficient of agreement for nominal scales. *Educational and Psychological Measurement* **20**(1), 37–46 (1960)
 - [37] Krippendorff, K.: Computing Krippendorff’s alpha-reliability. Technical report, Annenberg School for Communication, University of Pennsylvania, Philadelphia, PA (2011). https://repository.upenn.edu/asc_papers/43
 - [38] Yang, S.-w., Chi, P.-H., Chuang, Y.-S., Lai, C.-I.J., Lakhota, K., Lin, Y.Y., Liu, A.T., Shi, J., Chang, X., Lin, G.-T., *et al.*: SUPERB: Speech processing universal performance benchmark. arXiv preprint arXiv:2105.01051 (2021) <https://doi.org/10.21437/Interspeech.2021-1775>
 - [39] Kumar, A., Raghunathan, A., Jones, R., Ma, T., Liang, P.: Fine-tuning can distort pretrained features and underperform out-of-distribution. arXiv preprint arXiv:2202.10054 (2022)
 - [40] Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W., *et al.*: LoRA: Low-rank adaptation of large language models. *ICLR* **1**(2), 3 (2022)
 - [41] Houlshby, N., Giurgiu, A., Jastrzebski, S., Morrone, B., De Laroussilhe, Q., Gesmundo, A., Attariyan, M., Gelly, S.: Parameter-efficient transfer learning for NLP. In: International Conference on Machine Learning, pp. 2790–2799 (2019). PMLR
 - [42] Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., *et al.*: PyTorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems* **32** (2019)
 - [43] Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., Cistac, P., Rault, T., Louf, R., Funtowicz, M., *et al.*: Transformers: State-of-the-art natural language processing. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, pp. 38–45 (2020). <https://doi.org/10.18653/v1/2020.emnlp-demos.6>
 - [44] Taimoor Hassan, M., Ahmed, J., Awais, M.: Qalb: Largest state-of-the-art Urdu large language model for 230M speakers with systematic continued pre-training. arXiv e-prints, 2601 (2026)
 - [45] Lundberg, S.M., Lee, S.-I.: A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems* **30** (2017)

- [46] Poria, S., Cambria, E., Bajpai, R., Hussain, A.: A review of affective computing: From unimodal analysis to multimodal fusion. *Information Fusion* **37**, 98–125 (2017)
- [47] Lei, T., Barzilay, R., Jaakkola, T.: Rationalizing neural predictions. In: *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pp. 107–117 (2016)